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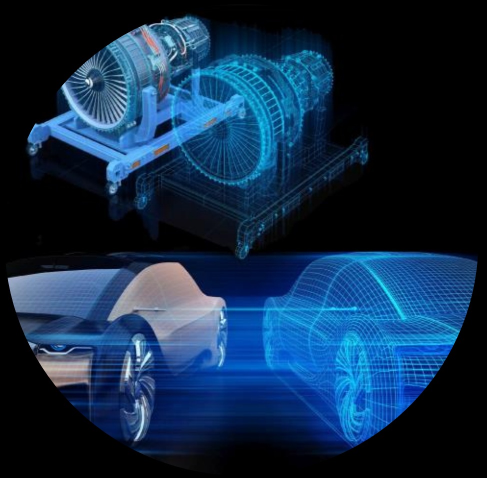
EVENT AND DATE

LECTURE 4 – DESIGN SKILLS

CS39A0040 Product & Service Development



Ilkka Donoghue, Post-Doctoral Researcher, ilkka.donoghue@lut.fi

AGENDA FOR LECTURE

- Guest Lecture – Antti Sääksvuori, Krios
- Design Skills

LUT University

> ” If you can’t do it with brainpower, you can’t do it with manpower—overtime, ”

Quote from Clarence L. Johnson (Kelly), Lockheed Martin

DESIGN SKILLS - THE BIG PICTURE

What are the skill needed in an engineer?

THE IMPORTANT SKILL (METHODS)

1 Planning	2 Discovering	3 Creating	4 Representing	5 Modelling
6 Prototyping	7 Experimenting	8 Evaluating	9 Deciding	10 Conveying

Refer to Development Reference Part II of the book

- Skills are *required* to evolve the design through the stages of development.
- Can be used in all phase / stages of development
- Drive the evolution of the development & design
- Expertise in the areas also define the designer or organisation
- Design tools & methods have been developed to help the designer to acquire & refine the skills.

IDENTIFY WHERE DIGITAL TOOLS ARE KEY?

What role do design applications (CFD, CAD, FEA/FEM, Simulation) and PLM play?

Planning can be formal and informal

SKILL 1 – PLANNING (FORMAL & INFORMAL)

- Planning is needed continuously throughout the development phases & processes

- Addresses the following area that must be considered:
  - Why is a design effort needed?
  - What resources are needed to enable the design?
  - What constraints limit the design
  - Information to guide actions must be effective at achieving the desired outcome & efficient use of resources.

The designer Point of View (PoV)

SKILL 1 – PLANNING

» For the designer, the Planning phase creates the information and insight

The purpose and objective of a project or task.

The starting and ending point of the project or task

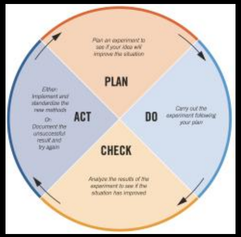
The constraints of the project or task

The resources available to carry out the project or task

The schedule related to the project or task.

Methods and Tools used in this planning (*Book Part II*):

- Critical Path Analysis
- Goal Pyramid
- Nucor's Circle
- Objective Tree
- Plan-Do-Check-Act (PDCA Cycle)
- Planning Canvas
- Project Objective Statement
- Value Engineering



Discovery Phase

SKILL 2 - DISCOVERING

- The Discovery phase is the most critical for the success of the product, service or Product-Service System.
- The Discovering (Discovery) is used to identify and define:
  - existing knowledge and ideas.
  - customer needs and preferences.

- Faster and less risks to build on existing knowledge and designs (Library vs laboratory)
- Reuse of existing Data, Information, Knowledge and Wisdom (DIKW)

Discovery Phase

SKILL 2 - DISCOVERING

» For the designer, the Discovery phase enables them to increase their knowledge:

Knowledge is limited or inadequate to be successful

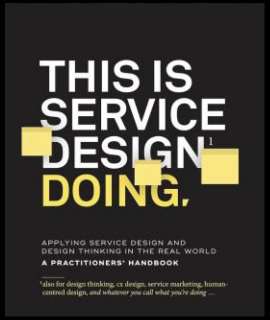
Knowledge is gained though persistence & creative exploration

Information needed exists, but must be found

Build on existing information (Digital Twins, PLM)

Existing innovations, technology, PSS and patents

Methods and Tools used in this planning (*Book Part II*):



Develop discovery skills

- Catalogue Search
- Codes & Standards (e.g. ISO)
- Delphi Method
- Internet Research

Uncover market needs & requirements

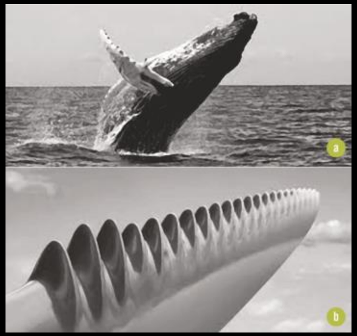
- Focus Groups
- Interviews, Observational Studies
- Surveys
- TRIZ (Theory of Inventive Problem Solving)

SKILL 3 - CREATING

- The Create Phase is where the Designer and/or Project Team
  - generating new knowledge
  - combining old knowledge in new and novel ways.
- Significant focus is on this area, but it is dependant of all other areas to bring significant value
- Key actions are organizing, imagining, formulating, brainstorming, synthesizing, substituting, and conceiving.

Create Phase

SKILL 3 - CREATING



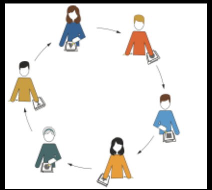
For the designer, the Create phase enables them to:

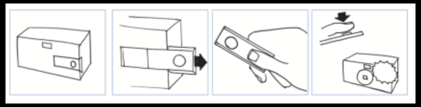
- Suspend judgment as well as apply judgment to ideas
- Risk of rejecting promising ideas as poor due to lack of DIKW
- Promising ideas must have had the chance to be revised & improved
- **Judgement must be applied to ensure quality of creation process outcomes**
- **Weaknesses must be identifies in the idea to avoid bad designs**

Methods and Tools used in this planning (*Book Part II*):

Develop discovery skills

- Bio-inspired Design
- Brainstorming
- Concept Classification Tree
- Decomposition
- Method 365
- Mind Maps
- Recombination Table
- SCAMPER
- Storyboards
- TRIZ





SKILL 4 - REPRESENTING

- The Representing Phase is where the Designer and/or Project Team create a graphical representations of the product to communicate the design to others
  - explain the idea of the design
  - define the geometry & other physical characteristics of the product (functions)
  - promote the product to secure approval

- There are multiple ways and tools to graphically represent the design:
  - Sketching communicates the intent of the design both within and without the team
  - Engineering drawings are precise, accurate and in scale being the major part of the design information
  - Solid Modelling creates a virtual model having identical geometry to the eventual physical part
  - Presentation Graphics are graphical representations prepared to communicate the design to people outside the development team to a non-technical audience

SKILL 4 - REPRESENTING

- For the designer, there are the following ways to represent the idea:

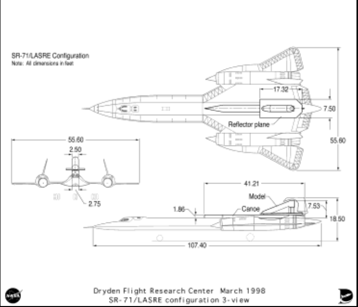
Sketching is for quick 2D or 3D representation of the design

Engineering drawings can be used early in the product development process and refined through time

Solid modelling are used for analysis of mass, FEM, CFD & real-time simulation modelling, and can generate Engineeting Drawings

Presentation graphics should reflect the quality of the design because the create the buy-in.

Methods and Tools used in this planning (*Book Part II*):



SKILL 5 - MODELLING

- The Modelling Phase is where the Designer and/or Project Team:
  - Can predict the performance of the design
  - Rapid iterations to the model can be done to refine the design so it achieves the performance criteria
  - Engineering models are validated to match reality and they enable cost effective design process
  - Reduce the reliance on series of prototypes.

- Real-time simulation based digital twins have proven their value to reduce development time, cost and risk.
- Models can be used in other areas of the product lifecycle

Modelling Phase

SKILL 5 - MODELLING

» The designer should in the modelling phase:

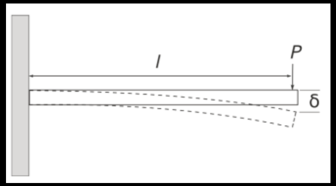
Evaluate the need to create multiple models and what type of modelling is needed to develop the design.

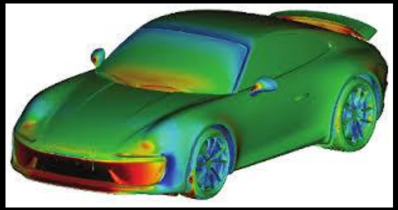
Early models will be rough, but will be refined over time as more insight is gained

Modelling should can include many types of models if they serve a purpose

Methods and Tools used in this planning (*Book Part II*):

- Design of Experiments
- Dimensional Analysis
- Finite Element Modelling
- Sensitivity Analysis
- Uncertainty Analysis



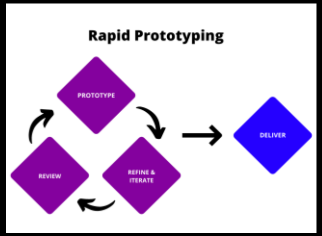


SKILL 6 - PROTOTYPING

- Creates physical approximations to the product, based on the current design, to help answer key questions about product
- Key actions and skills are, e.g., mock-ups, machining, casting, 3D printing, and breadboarding
- Effective prototypes are valuable (NOTE: Real-time simulation)

Methods and Tools used in this planning (*Book Part II*):

- Prototyping
- Rapid Prototyping



SKILL 7 - EXPERIMENTING

- Experimenting is where the Designer and/or Project Team use models, prototypes, and people to measure how the product should and does work.
  - Plan experiments
  - Test at multiple conditions
  - Complete all non-destructive tests before potentially destructive tests
  - Record all data and setup information necessary to perform the experiment
  - Calculate preliminary results during experimentation
  - Analyse the data statistically
  - Prepare a written summary with at least one summary data plot

- Key actions include setting up experiments, instrumenting test setups, statistical analysis, and working with human subjects.
- Formal experimentation is costly and must be planned, but give the best insight
- Informal experimentation is fast and cost effective ("*just trying the design out"*) to get insight of something working or not.

SKILL 8 - EVALUATING

- The Evaluating phase reviews the complete or part of the product by:
  - integrating the existing information to determine the quality of a design
  - Product desirability and transferability through-out product development for production and market launch
- Key actions are comparing, applying judgment, evaluating objectively, and concluding the feasibility
- Evaluation types through the produce development process are:
  - Intuition-based evaluation
  - Qualitative evaluation
  - Quantitative evaluation

SKILL 8 - EVALUATING

- The Evaluating types used in the different product development phases:

Opportunity Development	Concept Development	Subsystem Engineering	System Refinement	Producibility Refinement	Post-Release Refinement
Cost-benefit Analysis	Objective Tree	FEM	Fault Tree Anaysis (FTA)	Cpk, Cp analysis	Desing Experim ents
Multi-voting	Multi-voting	CFD	Sensitivity Analysis	Process FMEA	FMEA
	Scoring & Screening matrices	Desing Experiments			Cpk, Cp analysis
	Feasibilty Judgement	Failure Modes & Effects Analysis (FMEA)			Process FMEA
	Axiomatic Desgn	Axiomatic Desgn			

SKILL 9 - DECIDING

- The Decision phases is choosing a course of action based on the currently available information.
  - It is the act of advancing the design through decisions increasing the detail level captured in the design
  - In a production-ready product, every design detail is defined so a third party can manufacture identical products based on the design.
  - Avoiding or postponing decisions is avoiding or postponing progress and can cause extra costs
- Key actions are choosing and committing the design.

Decide Phase

SKILL 9 - DECIDING

» The Decision guidelines for the designer are:

Mundane & Vital Decisions:

The Vital Few

- are 10% - 20% decision with strong influence on how the design meets the market
- consume 80% - 90% of the design time.
- needs optimal solution

Mundane Many

- are 80% - 90% of the decisions
- Have minor influence on design
- Satisfactory solution

Motivation for Decision Making

- Decision making is a human activity
- Tools and methods are there to help make decisions
- Three levels of motivations:
  1. Individual Motivations
  2. Team Motivations
  3. Organizational Motivations

Decision Making Process:

Implicit and explicit methods

- Common process for implicit decions is not defined and there are pitfalls associated with implicit processes
- Explicit decision methods require the team to decide how a decision is to be made.
- there are two fundamental methods: voting and consensus.
- The Evaluation Matrix is used to reach a consensus

Decide-to-Decide

The key to effective progress in product development.

- Decision are vital for progress
- Be aggressive about making decisions.
- Decide, test, and validate.

Stakeholder buy-in is critical for the transition from design to production

SKILL 10 - CONVEYING

- Conveying is a course of action sharing the available information.
  - To meet the needs of others
  - advance the design,
  - facilitating appropriate evaluation by people both inside and outside the team.
- Key actions include salesmanship, writing, speaking, presenting, summarizing, and advocating.
- It is the responsibility of the design team to convey their successes (and concerns) to those outside the team who are stakeholders in the development project
  - understand audience needs
  - prepare effective communication materials to meet those needs
  - Benefits and limitations of the product must be transparent
  - Resources are commited based on stakeholder understanding of the benefits, risks and cost of the product


ANY
QUESTIONS
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